

HR Analytics — Employee Attrition Analysis

Introduction

Employee attrition refers to the gradual reduction of a company's workforce due to employees leaving and not being replaced. High attrition rates cost companies significant time and money in recruitment and training. This project aims to analyze employee attrition patterns and predict which employees are likely to leave using machine learning techniques.

Abstract

This project uses the IBM HR Analytics dataset containing 1470 employee records and 35 features. Exploratory Data Analysis (EDA) was performed to identify patterns. A Decision Tree classification model was built achieving 82.65% accuracy. SHAP analysis was used to identify the top factors driving attrition. An interactive Power BI dashboard was created to visualize findings.

Tools Used

- Python — Data cleaning, EDA, and machine learning
- Pandas & NumPy — Data manipulation
- Seaborn & Matplotlib — Data visualization
- Scikit-learn — Machine learning models
- SHAP — Model explainability
- Power BI — Interactive dashboard
- Jupyter Notebook — Development environment

Steps Involved

1. Loaded and explored the IBM HR Analytics dataset
2. Cleaned data by removing irrelevant columns
3. Performed EDA with 5 visualizations
4. Encoded categorical variables using Label Encoding
5. Built Logistic Regression (89.46%) and Decision Tree (82.65%) models
6. Applied SHAP analysis to identify key attrition drivers
7. Built interactive Power BI dashboard with 4 visuals and 2 slicers

Key Findings and Conclusion

SHAP analysis revealed the top 3 factors driving employee attrition:

1. OverTime — Employees who work overtime are significantly more likely to leave
2. Stock Option Level — Employees with no stock options feel less invested
3. Marital Status — Single employees tend to leave more than married ones

The Logistic Regression model achieved 89.46% accuracy while the Decision Tree model achieved 82.65% accuracy in predicting employee attrition.

Companies should focus on reducing overtime, offering stock options, and improving work life balance to retain employees and reduce attrition costs.